

School of Science, Computing and Engineering Technologies



COS10025

Technology in an Indigenous Context Project

Final project reflection report

Project Title: Preserving Hmong Cultural Heritage in Ha Giang
Through Smart Technologies

Student Name: Do Khac Phi

Student ID: 105560253

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Declaration

I declare that this report is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature:

A handwritten signature in black ink, consisting of a series of loops and a final horizontal stroke with a small 'N' or similar mark at the end.

Part A:

Introduction (Project Description)

Group 2 project focused on preserving the cultural heritage of the H'Mông ethnic minority community residing in Hà Giang province, Vietnam. With around 290,000 inhabitants in 2019, the H'Mông group is a big ethnic minority whose intangible heritage such as language, music, oral tradition, and rituals, is at risk as a result of modernization, migration, and integrating to the modern world. The project answers this key challenge of cultural deterioration by designing digital and sustainable solutions that are respectful of Indigenous knowledge systems and suited for remote areas with poor infrastructure.

The overarching objective was to use technology to strengthen community strategies, encourage the H'Mông people to record, conserve, and promote their culture for generations and among social groups. Six unique solutions were designed by individual team members, which had different elements of preservation, ranging from digital stories and language translation software to virtual travels and 3D scanning of cultural heritage. These solutions were brainstormed with continuous consultation, cyclic design workshops, and community-based thinking that respects cultural practices and ensures long-term sustainability.

Recommended option to proceed

I believe that Design Idea 1: Sustainable Tourism is the best option to develop out of the six design concepts that have been put forth. This one offers a well-rounded, people-centered strategy that incorporates sustainability, economic empowerment, and cultural heritage preservation. Because it utilizes local resources like traditional homes and oral tradition, as well as the growing national emphasis on cultural tourism, it is especially well-suited for the H'Mông people of Hà Giang.

A GPS-enabled smartphone app that allows visitors to explore thoughtfully planned cultural trails with abundant multimedia content is at the core of this concept. A localized web-based eco-homestay reservation system, run by smart IoT technology and powered by renewable energy, complements this improved experience. These technologies support community pride and cultural benefit by generating direct income for local H'Mông families in addition to drawing in culturally aware tourists.

From a sustainable perspective, operations using IoT monitoring systems and solar-powered smart homestays are associated with lower long-term operating costs and a minimal environmental impact. By utilizing traditional materials and construction techniques, the design also applies to local building practices, maintaining cultural integrity. Since locals assist with content creation, management, and tour leadership, the concept's central tenet is community ownership, which makes the solution robust and scalable.

This concept has a lot of potential for the future. To create a network of Indigenous cultural tourism, it can be duplicated with other ethnic minority communities throughout Vietnam. New features like virtual reality exposure, built-in marketplaces, or educational resources for schools can also be added to the website. The solution can serve as a prototype for moral and sustainable Indigenous tourism with the right instruction and steady development of digital skills.

Access and equity, cultural sensitivity, affordability, and sustainable livelihoods are just a few of the evaluation criteria that this design excels at fulfilling. Additionally, it keeps the H'Mông people in charge of telling their stories and showcasing their spaces while empowering them as entrepreneurs and cultural ambassadors. These factors make Design Idea 1 the most straightforward and successful strategy for preserving H'Mông culture in the rapidly evolving modern world.

Part B: Project reflection

Group Work Reflection

Working on this project with Group 2 was both an exciting and challenging experience. My team members—Lê Tuấn Huy, Vũ Huy Chris, Nguyễn Đình Tùng Dương, Nguyễn Đăng Nhật Minh, Nguyễn Đăng Tuấn Minh, and I collaborated on various design ideas, each targeting a unique aspect of preserving the H'Mông cultural heritage in Hà Giang. From the outset, we were united in our goal but had diverse perspectives and strengths, which made collaboration dynamic and often inspiring.

During the early weeks (Week 2–Week 4), we spent time identifying the core problem of cultural erosion and analyzing the specific needs of the H'Mông community. What worked well was our initial brainstorming and open-minded discussions. For example, Lê Tuấn Huy's idea about sustainable tourism sparked immediate interest and helped us frame our thinking around cultural immersion. We used Google Meet effectively during these stages to organize thoughts and align on objectives.

One strategy that worked particularly well was assigning clear leads for each design idea based on our strengths and interests. I led the development of Design Idea 4, focusing on cultural education via media and satellite outreach. This division helped us stay focused while also learning from each other's designs. Our weekly meetings, especially around Week 6 to Week 9, were productive and often resulted in useful peer feedback. I found Nguyễn Đăng Tuấn Minh's input on digital storytelling particularly insightful, and it pushed me to think more about community involvement in content creation.

However, not everything went smoothly. One recurring issue was time management, especially balancing individual schedules with our group timeline. On several occasions, we struggled to meet internal deadlines. For example, during Week 8, we had planned to review each other's draft ideas, but only two members submitted theirs on time, which delayed our internal peer review session. Another issue was communication clarity—sometimes messages on Teams were misread or not responded to promptly, leading to confusion about tasks.

From my perspective, what could be improved is early agreement on communication norms and clear meeting agendas. While our group was respectful, I noticed that some feedback remained vague, and I would have benefited from more constructive, specific input during key phases like the score sheet justification in Week 10.

An event that stood out in terms of team organisation was during Week 5, when Chris initiated a shared timeline on Google Sheets. This allowed us to visualize deadlines, link files, and assign roles. It gave structure to our work and motivated better accountability.

Our meetings in Week 7 and Week 9 were especially effective. One reason was that we began implementing shared minutes, which helped prevent repetition and ensured progress. For instance, Minh's summary from Week 7 helped me refine the visual aspects of my idea, such as the mobile culture vans.

In terms of delivery of the project design ideas, I was especially impressed by Duong's embroidery preservation concept. His offline-first approach influenced my own thinking about access equity and was a model of culturally sensitive, low-cost design. During presentation preparation in Week 11, our collaboration became more intense, and we collectively reviewed each other's slides to ensure consistency and strong narrative flow.

The delivery of our innovation concept—especially Tuấn Minh's digital storytelling platform—helped solidify the emotional impact of our project. It was in that phase that I fully appreciated the diversity of tools and formats we could use to engage both Indigenous and external audiences.

Finally, the delivery of our final presentation was a moment of pride. Even though we had different speaking styles, we supported each other well, and I believe our presentation reflected genuine team effort and cross-learning.

Individual Work Reflection

In this project, I was responsible for leading the development of Design Idea 4: Cultural Exposure Through Media and Education. My aim was to create inclusive educational and media tools—such as satellite classes, mobile cultural vans, and national TV programming—that would promote H'Mông traditions in both rural and urban areas. This design aligned with my interest in storytelling and community-based outreach, especially in regions with limited internet connectivity.

Project Tasks by Phase

In Phase 1, I contributed to identifying the key issues facing the H'Mông community, particularly around the decline of language use, oral storytelling, and ritual practices. During Week 3's team brainstorming, I suggested that media and education could form a cultural bridge between older generations and youth. I conducted background research using sources such as Vietnam Discovery (2021) and UN Tourism (2024) to understand the viability of satellite broadcasting and mobile vans in mountainous regions like Hà Giang.

In Phase 2, I began designing my solution. I drafted the initial structure of the three-pronged approach—TV programs, satellite e-classes, and culture vans—and shared it during our Week 5 workshop. I refined the proposal using team feedback. I also evaluated how this idea could complement other members' designs, such as Minh's translation app or Huy's tourism platform. By Week 6, I had created a visual outline showing how educational content could be shared across multiple formats.

During Phase 3, I completed the score sheet assessment by applying the six COS10025 guidelines. My design scored highest in "Access & Equity" and "Cultural Appropriateness" due to its multimedia format and inclusion of elders in content creation. I took care to evaluate limitations honestly, such as cost and training needs, and presented them during our Week 9 peer review session. This feedback helped me adjust the scope and refine which technologies were most realistic.

In Phase 4, I revised my design by integrating more hands-on elements. Based on Duong's suggestion, I added community-led workshops and DVD distribution to the mobile vans. These changes helped ensure that the content would remain useful even without strong internet or digital devices.

Contributions to the Group

Throughout the project, I regularly shared progress in our Google Meet group and offered feedback on other members' drafts. I helped compile research resources, particularly academic and government sources about H'Mông culture, which were used by multiple members. I also coordinated with Chris to ensure our references and APA formatting were consistent in the final document.

In terms of group dynamics, I participated actively in team discussions and tried to foster a supportive environment. I encouraged quiet members to contribute and helped facilitate some of our weekly meetings when the assigned facilitator was unavailable. I learned how important patience and empathy are in teamwork—especially when working with different personalities and communication styles.

Conclusion and Recommendation

Looking back, I believe I contributed a culturally appropriate and scalable design idea that complements the broader goal of preserving H'Mông heritage. I am proud that my proposal considered both technological possibilities and cultural sensitivities, especially by placing elders and youth at the center of the solution.

In future group work, I hope to improve by managing my time more efficiently and seeking feedback earlier in the process. I also want to develop stronger visual presentation skills to better communicate my design ideas. If I were to expand on this project, I would like to explore partnerships with local educators, NGOs, or media outlets to test pilot programs in remote schools and villages.

Part C: Unit Learning Outcomes (ULOs)

ULO 1: Locate Indigenous knowledge systems and consider how they story the long history of technology, science, and engineering.

In the early phase of the project (Week 2–Week 4), I explored how Indigenous knowledge systems, particularly those of the H'Mông community, have long existed in the form of oral storytelling, textile design, and agricultural practices. Our team spent time researching traditional practices such as rammed-earth house construction and the Gầu Tào festival. My task was to identify how these traditional knowledge systems could be communicated to a younger generation in an engaging and accessible way. I proposed using media formats like television, satellite classes, and mobile vans to present this knowledge, with elders as the primary narrators. The result was a culturally sensitive design that used modern formats to preserve ancient wisdom. This showed me that Indigenous knowledge is not outdated—it is simply under-acknowledged by Western systems.

ULO 2: Explain the importance of, and find opportunities to, respectfully converge Western knowledge systems with Indigenous knowledge systems.

During the development of Design Idea 4, I had to find ways to integrate broadcasting, digital content, and satellite infrastructure—technologies based in Western systems—with Indigenous values of oral transmission and collective memory. My task was to create a respectful bridge between these two systems. I did this by ensuring that content development would be led by local elders and educators, not imposed externally. I also designed tools like offline DVDs and print materials, ensuring low-tech options were included. The result was a design that encouraged respectful convergence: modern tools supported traditional learning rather than replacing it. This helped me understand that technology can empower Indigenous systems, but only if applied in ways that respect community ownership and knowledge sovereignty.

ULO 3: Apply relevant knowledge of emerging technologies to a project within an Indigenous context, taking into consideration and acknowledging Indigenous histories, worldviews, standpoints, and cultures.

In my role, I applied knowledge of educational broadcasting, community workshops, and hybrid media delivery to preserve the H'Mông cultural identity. I recognized early on that limited internet access in Hà Giang would make purely digital solutions less effective. My task was to adapt emerging technologies like satellite broadcasting and mobile vans into a format that worked in remote areas. I researched existing use cases from Vietnam and adapted them for cultural education. I also ensured that every tool was grounded in H'Mông worldviews—where elders are cultural authorities and knowledge is passed orally. The result was a multi-channel delivery model that could promote cultural pride, preserve language, and bridge generational gaps. I learned that technology must follow cultural logic, not the other way around.

ULO 4: Function as an effective team member using project management tools and demonstrating professionalism and ethical behaviour.

Throughout the project, I worked to be a supportive and consistent team member. Our group used Microsoft Teams, Miro, and Google Sheets to manage timelines and resources. My task was to contribute my own work while also helping others stay on track. I posted regular updates, responded to feedback, and helped standardize our APA references. One specific moment that demonstrated ethical behaviour was during our Week 7 team check-in, when I noticed a teammate's design could potentially misrepresent a sacred H'Mông ritual. I encouraged that member to seek community input and revise the framing, which they did. The result was a stronger, more respectful final product. This taught me that teamwork involves not only doing your part, but helping others do theirs ethically and thoughtfully.

**Word count (not included headings, declaration and acknowledgement):
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